LEARNER GUIDE

Percentage change

Percentages, Ratio, Growth and Bounds · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **percentage change** compares the change with the **original** (starting) value: to find it, divide the change by the original and multiply by 100. A rise is a percentage increase and a fall is a percentage decrease. The new value divided by the original is the multiplier: 1.25 means a 25% increase, 0.8 a 20% decrease. The same change can be a big percentage of a small amount and a small percentage of a large one.

What you are learning to notice

- Always compare with the original: the value before the change.
- £80 → £100 is +25%, but £100 → £80 is –20%: the same £20, different originals.
- A bigger change in pounds is not always a bigger percentage change.
- The new value divided by the original is the multiplier from Lesson 3.
- A decrease cannot be more than 100%, but an increase can.

Moving through it, one step at a time

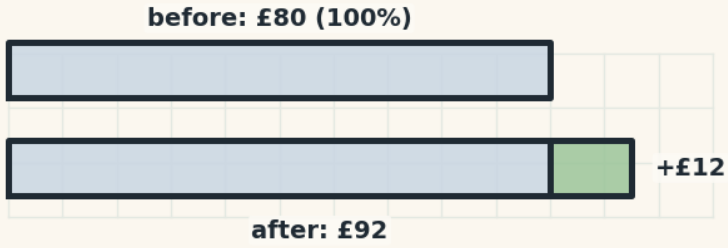
Find the change: new – original. Divide by the original. Multiply by 100. Say whether it is an increase or a decrease. Or: divide the new value by the original, and read off the multiplier (1.15 is +15%, 0.85 is –15%).

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A percentage increase

A jacket cost £80 last year and costs £92 now. What is the percentage increase?

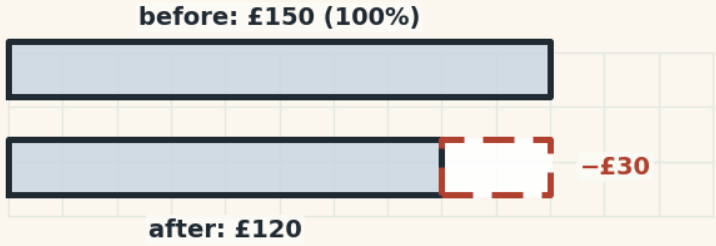


1. The change	£92 – £80 = £12
2. Compare with the original	$\frac{12}{80} = 0.15$
3. As a percentage	0.15 × 100 = 15%
4. Check with the multiplier	$\frac{92}{80} = 1.15$

A 15% increase.

Example 2 — A percentage decrease

A phone falls in price from £150 to £120. What is the percentage decrease?



1. The change	£150 – £120 = £30
2. Compare with the original, £150	$\frac{30}{150} = 0.2$
3. As a percentage	20%

A 20% decrease (not 25%, which would compare with the new price, £120).

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (percentages as multipliers (Lesson 3) and one number as a percentage of another), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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